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IPV6 pour les réseaux IoT

1. IoT network characteristics and specificities

Hint : *List the major peculiarities of IoT physical networks. If needed, you can take the case of Low-Power Wireless Personal Area Networks (LP-WPAN) that we considered during the course and explain how they differ from conventional computer networks and what are the specific constraints that they are subject to.*

IoT physical networks, especially Low-Power Wireless Personal Area Networks (LP-WPAN), differ significantly from traditional computer networks. They are designed for low power consumption, limited bandwidth, and short-range communication, often operating in ad-hoc or mesh topologies. Devices in these networks are resource-constrained, with limited memory, processing power, and battery life, which necessitates lightweight protocols and energy-efficient operations. Unlike conventional networks, LP-WPANs prioritize scalability and reliability in harsh or dynamic environments, where links may be unstable. The IEEE 802.15.4 standard, for example, defines the physical and MAC layers for such networks, emphasizing low data rates and minimal energy use.

2. Rationale for adopting an IPv6 based architecture to support the communications of an IoT system or use case

Hint : *List the main benefits of adopting an IP based architecture in an IoT system, up the connected object (e.g. sensor, etc.).*

An IPv6-based architecture is ideal for IoT systems due to its vast address space, which ensures every device—from sensors to actuators—can have a unique global address. This eliminates the need for NAT and simplifies end-to-end communication. IPv6’s stateless address autoconfiguration (SLAAC) enables plug-and-play integration, reducing manual configuration efforts. Built-in support for IPsec enhances security, while features like multicast and anycast optimize data distribution and service discovery. For IoT applications, IPv6’s mobility protocols ensure seamless connectivity for moving devices, making it a future-proof choice for smart cities, industrial automation, and healthcare systems.

3. IPv6 basics

Hint : First, from the experiments and traffic captures that you did during TD1, describe the different IPv6 initialisation steps that a host goes through, when switched on. Explain the rationale of the different steps, and the messages (with the types of IPv6 addresses) that are used to complete these steps. Then, derive some of the requirements of IPv6 (in terms of transmission capabilities of the physical network, and host availability) and enrich them with some other important characteristics of IPv6.

When an IPv6 host powers on, it undergoes several initialization steps. First, it generates a link-local address using its MAC address (EUI-64 format) and verifies its uniqueness via Duplicate Address Detection (DAD). The host then sends a Router Solicitation (RS) message to discover routers and obtain network parameters, such as a global prefix. Combining this prefix with its interface ID, the host configures a global IPv6 address. Neighbor Discovery Protocol (NDP) messages, like Neighbor Solicitation (NS) and Neighbor Advertisement (NA), help maintain link-layer connectivity.

IPv6 requires a minimum MTU of 1280 bytes to avoid fragmentation, which is critical for constrained IoT links. Multicast support is essential for efficient service discovery, and hosts must balance energy efficiency with network availability, often using sleep modes to conserve power.

4. IPv6 adaptation and extensions in order to enable its use atop a physical IoT network

Hint :Without delving into the details, and relying on the experiment that you undertook during TD2, list the main additions, adjustments and optimizations of IPv6 that were defined for an application in the context of an IoT network.

To operate over IoT physical layers like IEEE 802.15.4, IPv6 is adapted through protocols like 6LoWPAN. This protocol compresses IPv6 headers, reducing overhead from 40 bytes to as little as 2 bytes, and handles fragmentation for larger packets. Routing protocols like RPL (Routing Protocol for Low-Power and Lossy Networks) optimize path selection in mesh topologies, while mechanisms like TSCH (Time-Slotted Channel Hopping) in IEEE 802.15.4e enhance reliability and energy efficiency. These adaptations ensure IPv6 can function effectively in resource-constrained environments.

5. The IETF IPv6 based stack for IoT

Hint :Depict the protocol tack proposed by the IETF for IoT and then briefly describe the main network functions performed by the new layers. Also, provide a few words to describe the proposed application level protocols.

The IETF's proposed stack for IoT includes IPv6 at the network layer, with 6LoWPAN providing adaptation for constrained links. Above IPv6, lightweight application protocols like CoAP (Constrained Application Protocol) and MQTT-SN (MQTT for Sensor Networks) enable efficient communication. CoAP, a RESTful protocol, uses UDP for low overhead, while MQTT-SN supports publish-subscribe messaging in bandwidth-limited environments. RPL handles routing in lossy networks, ensuring robust connectivity for large-scale IoT deployments.

6. Existing IPv6 based network technologies for IoT

Hint : List the existing IoT network technologies that are using IPv6 and their associated vertical(s) (application domain(s))

Several IPv6-based technologies are widely used in IoT applications. **6LoWPAN** is prevalent in smart homes and industrial automation, while **Thread** powers devices like Google Nest and Apple HomeKit. **LoRaWAN** and **NB-IoT** leverage IPv6 for smart cities and asset tracking, and **BLE Mesh** enables IPv6-based communication in building automation. These technologies demonstrate IPv6's versatility across verticals, from healthcare to logistics.

7. Is an IPv6 based stack relevant for your semester project ?

Hint : After briefly describing your semester project, elaborate very shortly on the relevance of adopting IPv6 in your semester project.

Our project involves an intracranial pressure sensor system. Each sensor uses an NXP9080-DK board and communicates via BLE with a Raspberry Pi. There is one board per sensor.

In this context, adopting an IPv6-based stack is not particularly relevant. The system involves a limited number of sensors and relies on short-range BLE communication rather than IP-based networking. The sensors do not require direct Internet connectivity, routing, or large-scale addressing, which are the main advantages of IPv6.

Using IPv6 would introduce additional complexity and protocol overhead without providing significant benefits for this specific architecture.

Réseaux émergents

1. What makes SDN different from legacy computer networks ? What are the appealing opportunities that it paves the way for ? What are its main challenges ?

Hint : Shortly, list the SDN principles that do not hold for legacy computer networks. Briefly, elaborate on these principles to sketch some of the opportunities that SDN brings.

In traditional computer networks, each device such as a router or a switch makes its own decisions. The control logic and the packet forwarding are embedded in the same equipment, which makes the network hard to manage and difficult to evolve.

SDN takes a different approach by separating the control plane from the data plane. The network intelligence is moved to a centralized controller, while network devices are only responsible for forwarding traffic. This change makes the network easier to understand, configure, and adapt.

What makes SDN particularly interesting is that the network becomes programmable. Instead of manually configuring each device, network behavior can be modified through software, which opens the door to automation and faster innovation.

However, SDN also comes with challenges. The controller can become a bottleneck or a single point of failure if not properly designed. Security and scalability are also important concerns, especially in large deployments.

2. What does NFV (Network Function Virtualization) stand for ? What are the opportunities that it paves the way for ?

NFV stands for Network Function Virtualization. The main idea is to replace dedicated network hardware (such as firewalls, gateways or load balancers) with software-based functions running on standard servers.

This approach makes networks more flexible and cost-effective. Network services can be deployed faster, scaled on demand, and updated more easily. NFV is especially attractive in cloud, 5G and IoT environments, where flexibility and rapid deployment are key requirements.

3. Are SDN and/or NFV relevant for your semester project ? If not, choose one of the assignments below ?

Hint : After briefly describing your semester project, elaborate very shortly on the relevance of adopting SDN and/or NFV in your semester project. Alternatively, choose one of the papers below, which propose some concrete applications of SDN/NFV in the IoT context. Read the recommended sections and answer the corresponding questions.

Our semester project focuses on an intracranial pressure sensor. Each sensor is connected to an NXP9080-DK board and communicates via Bluetooth Low Energy with a Raspberry Pi acting as a central node.

In this context, SDN and NFV are not really relevant. The system is small, local, and does not rely on IP routing or complex network management. Introducing SDN or NFV would add unnecessary complexity without bringing real benefits. These technologies are better suited for large-scale IoT networks with many devices and dynamic traffic management needs.

Choix du document:: LoRa-SDN: Providing Wireless IoT Edge Network Functions via SDN

The authors propose using SDN in LoRa-based IoT networks to make them more flexible and easier to manage. In large deployments such as smart cities, traditional LoRa networks are mostly static and difficult to adapt to changing conditions.

By introducing an SDN controller, the network gains a global view of the system. This makes it possible to better control traffic, manage gateways more efficiently, and react dynamically to congestion or failures. The idea is to extend SDN principles beyond the core network and apply them at the IoT edge.

Overall, the proposed LoRa-SDN architecture shows how SDN can simplify the management of large-scale IoT networks while improving flexibility and control, something that would be much harder to achieve with a conventional LoRa architecture.